A. Tender Carpenter

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: dp, geometry, greedy, math | https://codeforces.com/problemset/problem/2053/A

I would use a firework to announce, a wave to bid farewell, and a bow to say thanks: bygones are bygones; not only on the following path will I be walking leisurely and joyfully, but also the footsteps won't halt as time never leaves out flowing; for in the next year, we will meet again.

— Cocoly1990, *Goodbye 2022*

In his dream, Cocoly would go on a long holiday with no worries around him. So he would try out for many new things, such as... being a carpenter. To learn it well, Cocoly decides to become an apprentice of Master, but in front of him lies a hard task waiting for him to solve.

Cocoly is given an array $a_1, a_2, \ldots, a_n$. Master calls a set of integers S *stable* if and only if, for any possible u, v, and w from the set S (note that u, v, and w do not necessarily have to be pairwise distinct), sticks of length u, v, and w can form a non-degenerate triangle[□].

Cocoly is asked to partition the array a into several (possibly, 1 or n) **non-empty** continuous subsegments[†], such that: for each of the subsegments, the set containing all the elements in it is *stable*.

Master wants Cocoly to partition a in **at least two** different[‡] ways. You have to help him determine whether it is possible.

[□]A triangle with side lengths x, y, and z is called non-degenerate if and only if:

- $x + y$ gt z,
- $y + z$ gt x, and
- $z + x$ gt y.

[†]A sequence b is a subsegment of a sequence c if b can be obtained from c by the deletion of several (possibly, zero or all) elements from the beginning and several (possibly, zero or all) elements from the end.

[‡]Two partitions are considered different if and only if at least one of the following holds:

- the numbers of continuous subsegments split in two partitions are different;
- there is an integer k such that the lengths of the k-th subsegment in two partitions are different.

Input

Each test contains multiple test cases. The first line of the input contains a single integer t ($1 \le t \le 200$) — the number of test cases. The description of test cases follows.

The first line of each test case contains a single integer n ($2 \le n \le 200$) — the length of the array a.

The second line contains n integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le 10^5$) — the elements in the array a.

Output

For each test case, print `YES` if there are at least two ways to partition a, and `NO` otherwise.

You can output the answer in any case (upper or lower). For example, the strings `yEs`, `yes`, `Yes`, and `YES` will be recognized as positive responses.

Example

Input
5 4 2 3 5 7 4 115 9 2 28 5 8 4 1 6 2 6 1 5 4 1 4 7 2 100000 100000

Output
YES NO NO YES YES

Note

In the first test case, here are two possible partitions:

- [2, 3], [5, 7], since
- [2, 3] is *stable* because sticks of lengths (2, 2, 2), (2, 2, 3), (2, 3, 3), (3, 3, 3) respectively can all form non-degenerate triangles.
- [5, 7] is *stable* because sticks of lengths (5, 5, 5), (5, 5, 7), (5, 7, 7), (7, 7, 7) respectively can all form non-degenerate triangles.
- and [2], [3, 5], [7], since
- [2] is *stable* because sticks of lengths (2, 2, 2) respectively can form a non-degenerate triangle.
- [3, 5] is *stable* because sticks of lengths (3, 3, 3), (3, 3, 5), (3, 5, 5), (5, 5, 5) respectively can all form non-degenerate triangles.
- [7] is *stable* because sticks of lengths (7, 7, 7) respectively can form a non-degenerate triangle.

Note that some other partitions also satisfy the constraints, such as [2], [3], [5], [7] and [2], [3], [5, 7].

In the second test case, Cocoly can only partition each element as a single subsegment, resulting in [115], [9], [2], [28]. Since we only have one possible partition, the answer is `NO`.

In the third test case, please note that the partition [8, 4], [1], [6], [2] does not satisfy the constraints, because {8, 4} is not a *stable* set: sticks of lengths 4, 4, and 8 cannot form a non-degenerate triangle.